



Artificial Intelligence Techniques



INTRODUCTION TO NEURAL NETWORKS 1



Aims: Section

fundamental theory and
practical applications of
artificial neural networks.



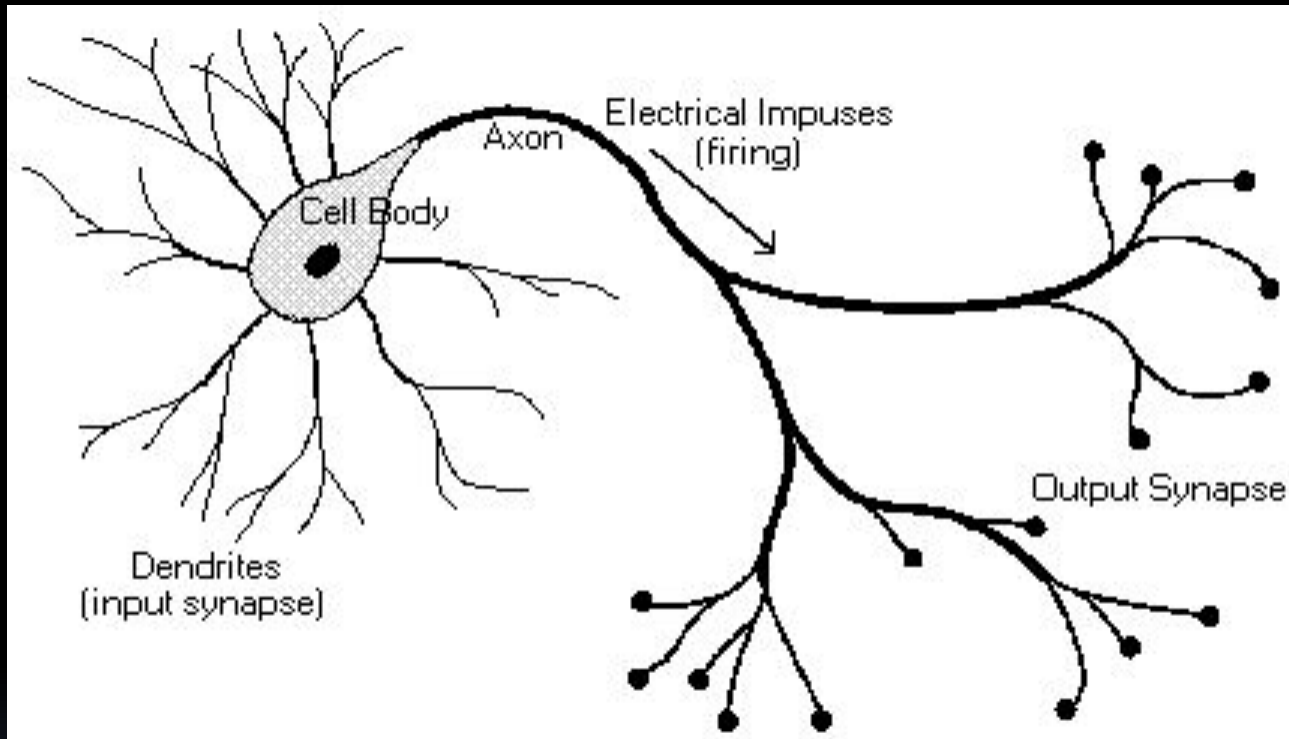


Aims: Session Aim

Introduction to the
biological background and
implementation issues
relevant to the
development of practical
systems.



Biological neuron



- Taken from <http://hepunx.rl.ac.uk/~candreop/minos/NeuralNets/neuralNetIntro.html>



Human brain consists of
approx. 10 billion
neurons interconnected
with about **10 trillion**
synapses .

- 
- A neuron: specialized cell for **receiving, processing and transmitting** informations.

- 
- Electric charge from neighboring neurons reaches the neuron and they add.
- 



□ The summed signal is passed to the soma that processing this information.



□ A signal threshold is applied.



If the summed signal $>$
threshold, the neuron fires





Constant output signal is
transmitted to other
neurons.

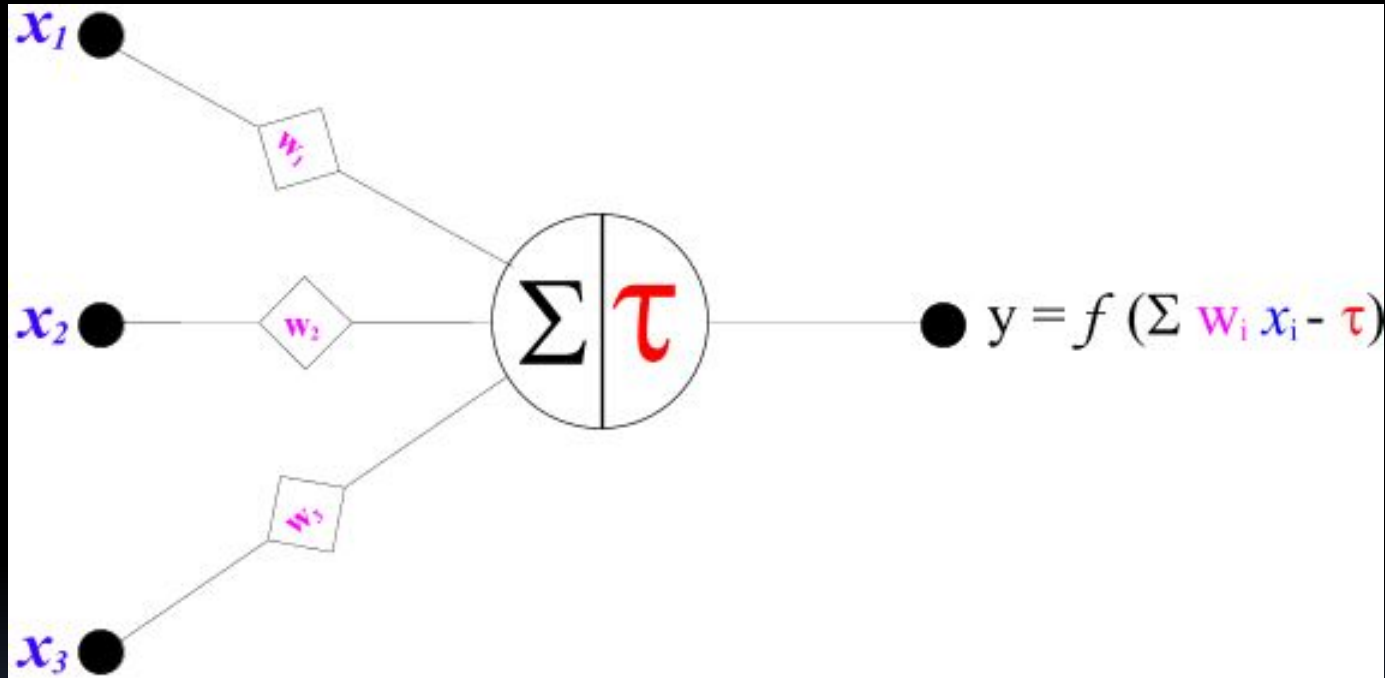


- 
- The strength and polarity of the output depends features of each synapse
- 

- 
- 
- varies these features - adapt the network.

- 
- varies the input contribute -
vary the system!
- 

Simplified neuron



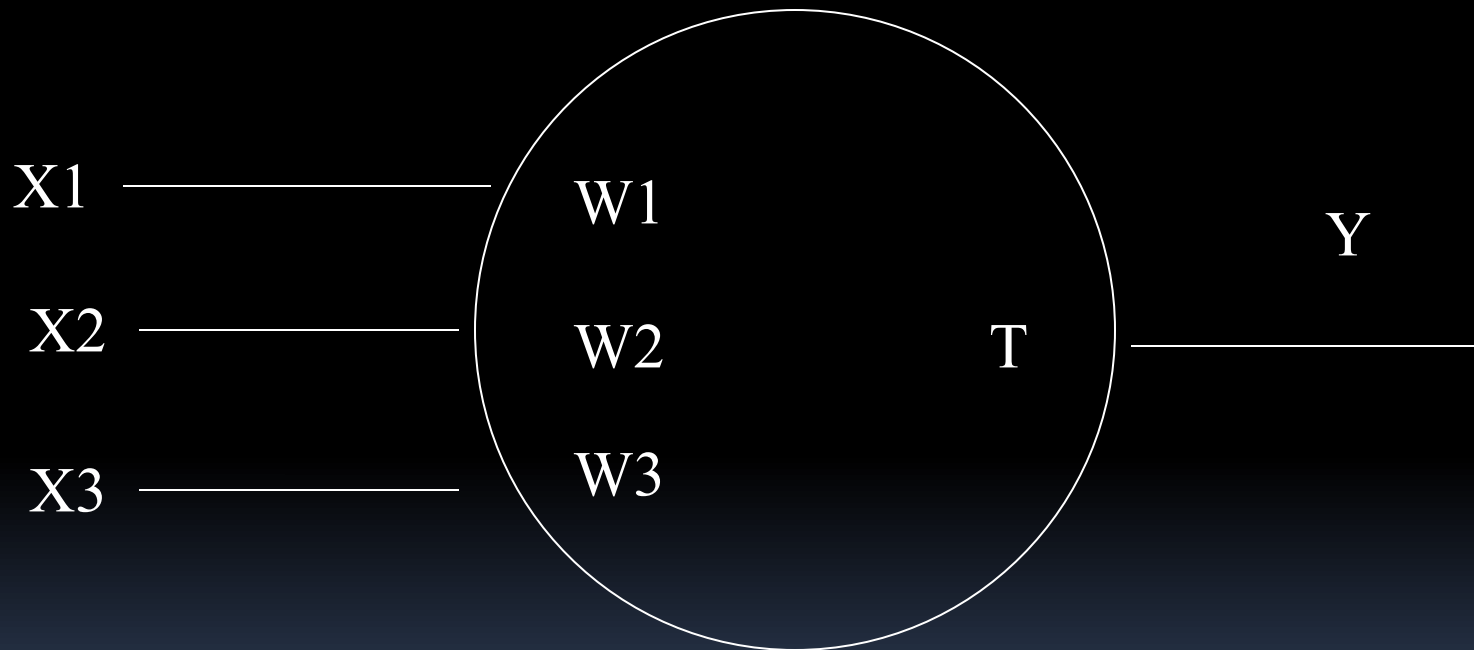
- Taken from <http://www.geog.leeds.ac.uk/people/a.turner/projects/medalus3/Task1.htm>



Exercise 1

- In groups of 2-3, as a group:
 - Write down one question about this topic?
- 

McCulloch-Pitts model



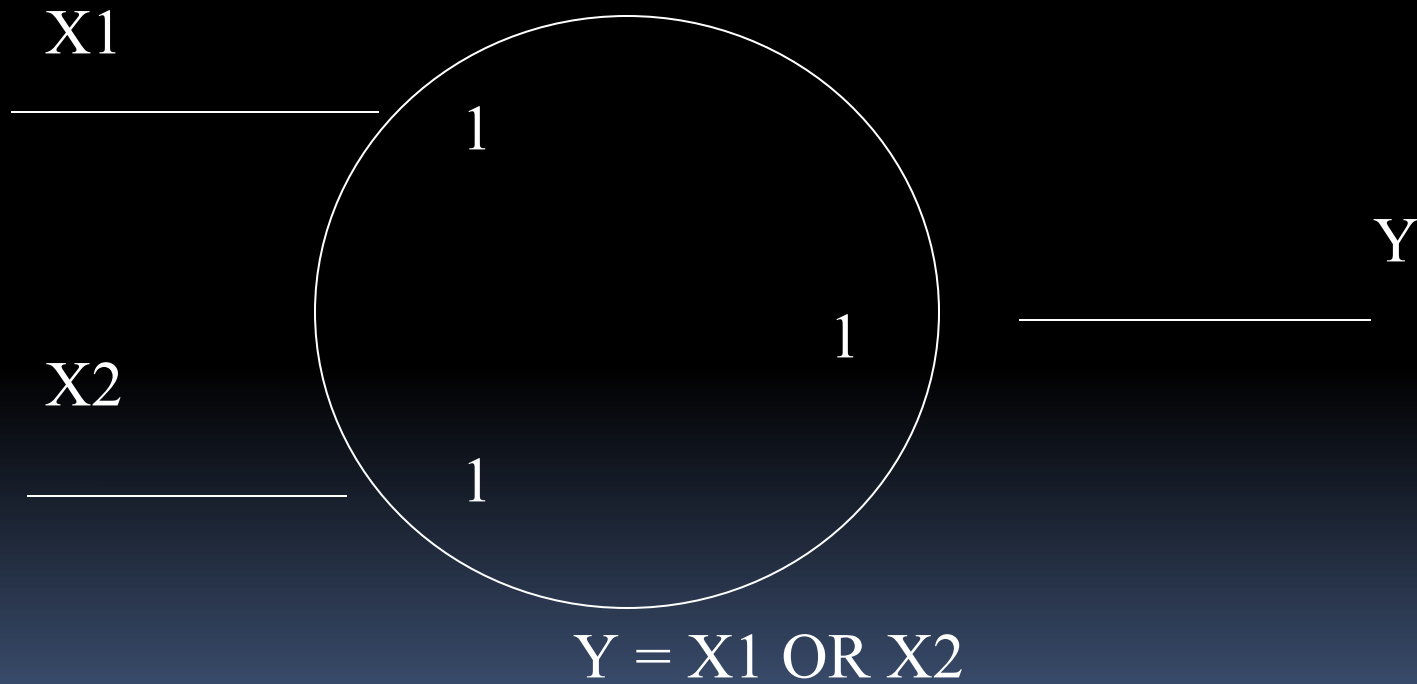
$Y=1$ if $W_1X_1+W_2X_2+W_3X_3 \geq T$
 $Y=0$ if $W_1X_1+W_2X_2+W_3X_3 < T$

McCulloch-Pitts model

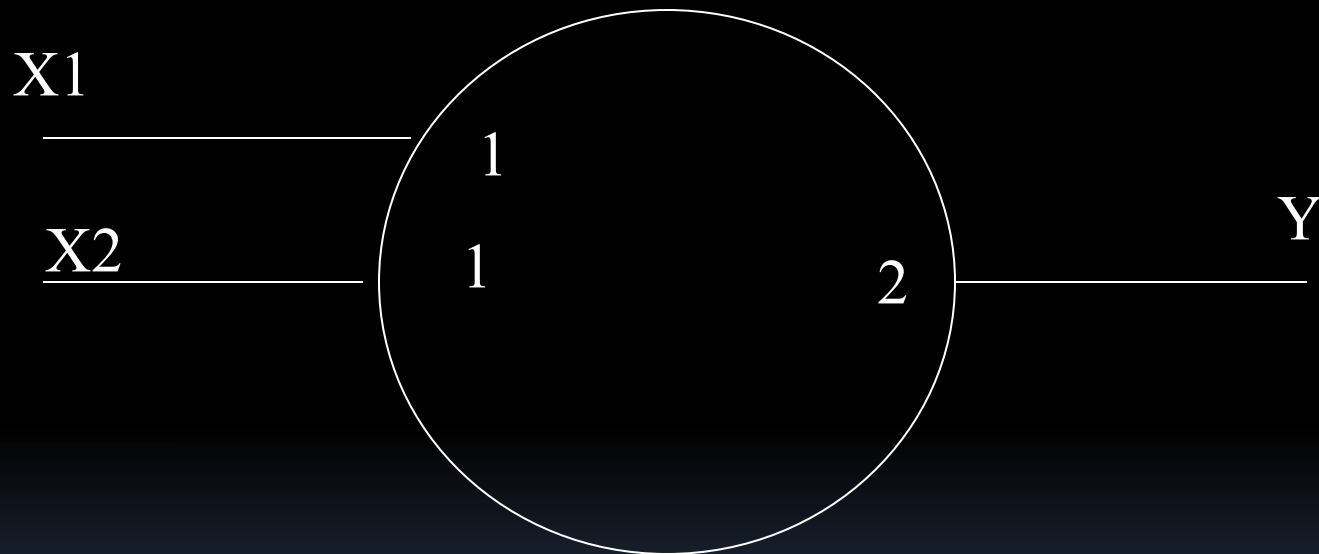
$Y=1$ if $W_1X_1+W_2X_2+W_3X_3 \geq T$

$Y=0$ if $W_1X_1+W_2X_2+W_3X_3 < T$

Logic functions - OR

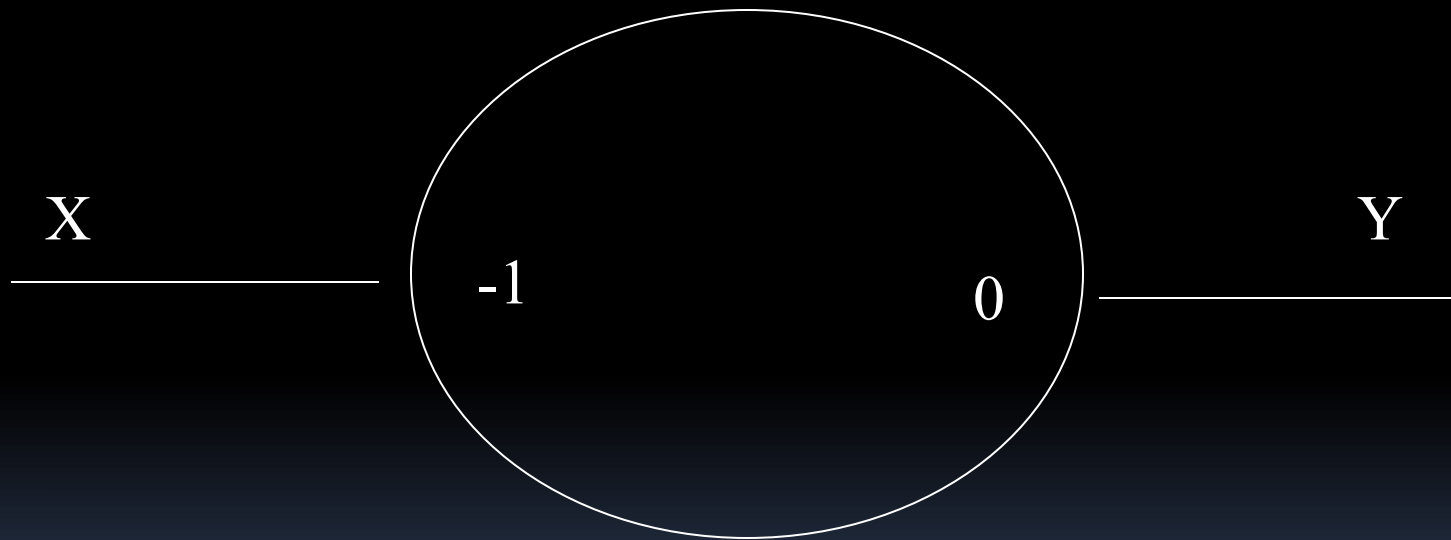


Logic functions - AND



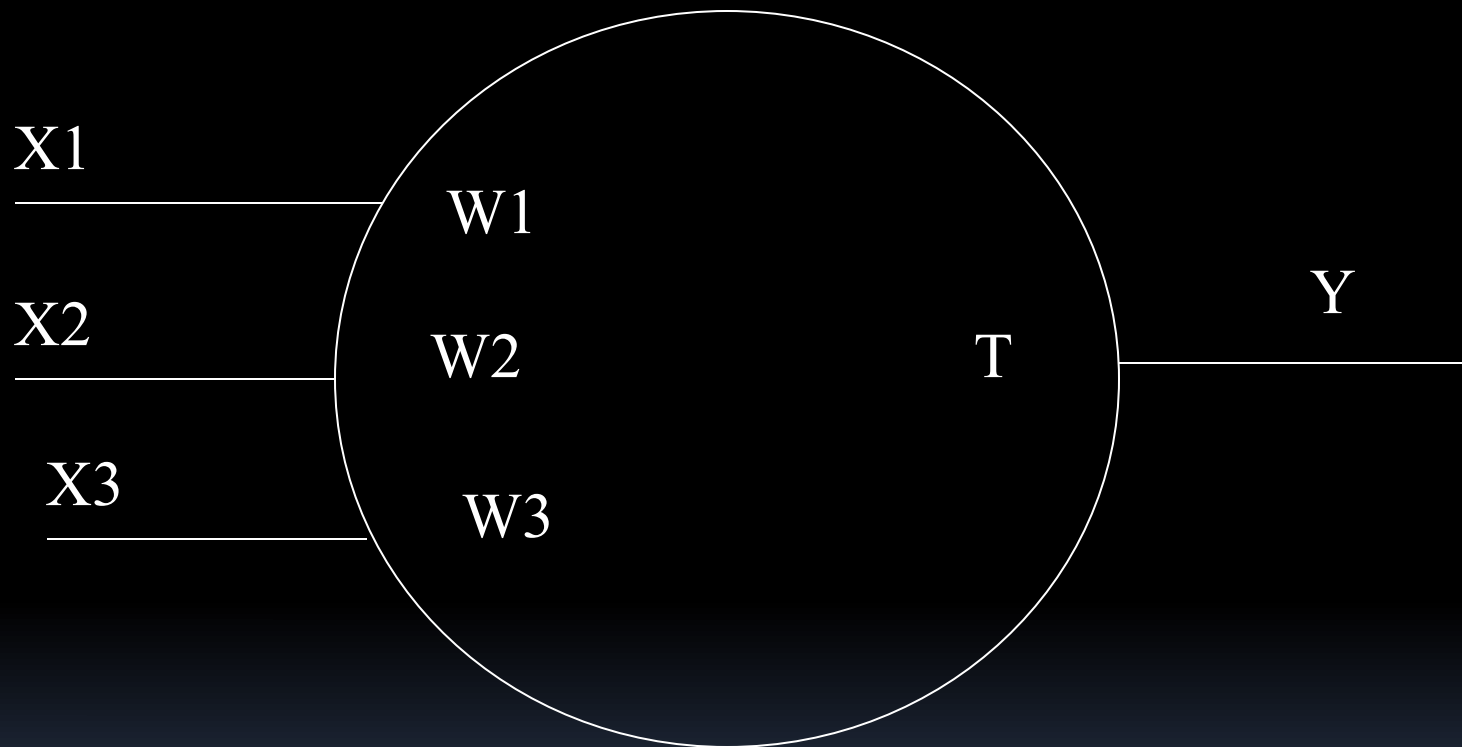
$$Y = X1 \text{ AND } X2$$

Logic functions - NOT



$$Y = \text{NOT } X$$

McCulloch-Pitts model



$Y=1$ if $W_1X_1+W_2X_2+W_3X_3 \geq T$

$Y=0$ if $W_1X_1+W_2X_2+W_3X_3 < T$

Introduce the bias

Take the threshold over to the other side of the equation and replace it with a weight W_0 which equals $-T$, and include a constant input X_0 which equals 1.

Introduce the bias

$$Y=1$$

$$\text{if } W_1X_1 + W_2X_2 + W_3X_3 - T \geq 0$$

$$Y=0$$

$$\text{if } W_1X_1 + W_2X_2 + W_3X_3 - T < 0$$



Introduce the bias

- Lets just use weights – replace T with a ‘fake’ input
 - ‘fake’ is always 1.
- 

Introduce the bias

$$Y=1$$

$$\text{if } W_1X_1 + W_2X_2 + W_3X_3 + W_0X_0 \geq 0$$

$$Y=0$$

$$\text{if } W_1X_1 + W_2X_2 + W_3X_3 + W_0X_0 < 0$$

Short-hand notation

Instead of writing all the terms in the summation,
replace with a Greek sigma Σ

$$Y=1 \text{ if } W_1X_1+W_2X_2+W_3X_3 +W_0X_0 \geq 0$$

$$Y=0 \text{ if } W_1X_1+W_2X_2+W_3X_3 +W_0X_0 < 0$$

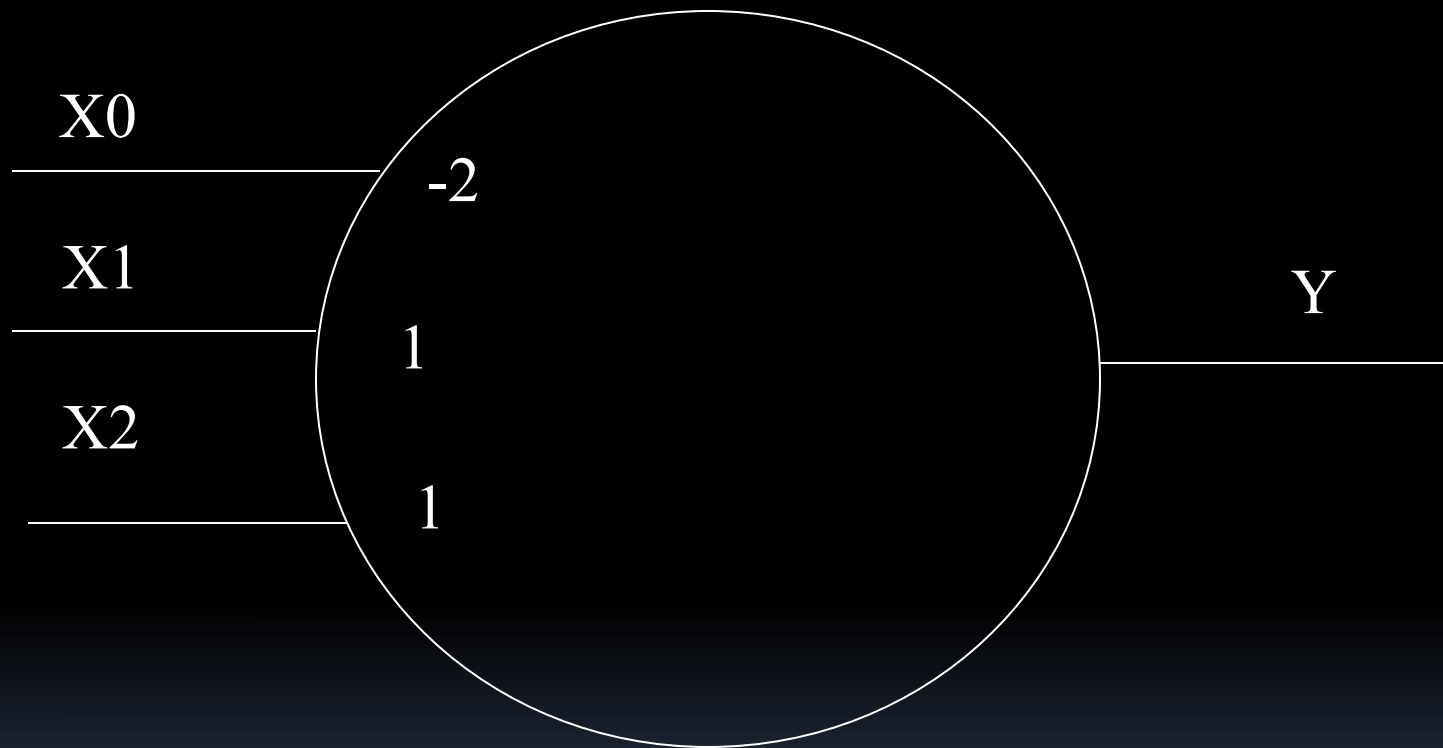
becomes

Logic functions - OR



$$Y = X1 \text{ OR } X2$$

Logic functions - AND



$$Y = X1 \text{ AND } X2$$

Logic functions - NOT



$$Y = \text{NOT } X1$$

The weighted sum

- The weighted sum, $\sum W_i X_i$ is called the “net” sum.

$$\square \text{Net} = \sum W_i X_i$$

$$\square y=1 \text{ if net} \geq 0$$

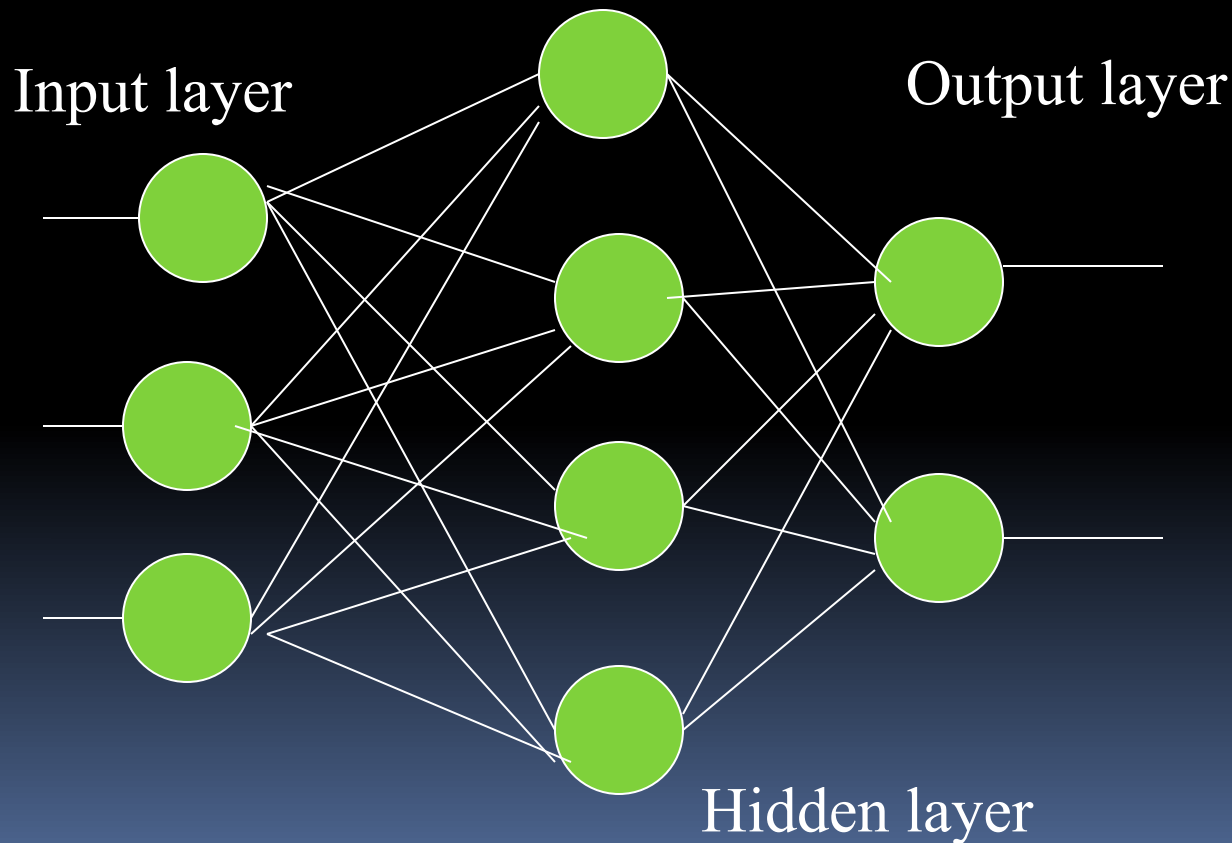
$$\square y=0 \text{ if net} < 0$$



Multi-layered perceptron

- Feedback network
 - Train by passing error backwards
 - Input-hidden-output layers
 - Most common
- 

Multi-layered perceptron (Taken from Picton 2004)

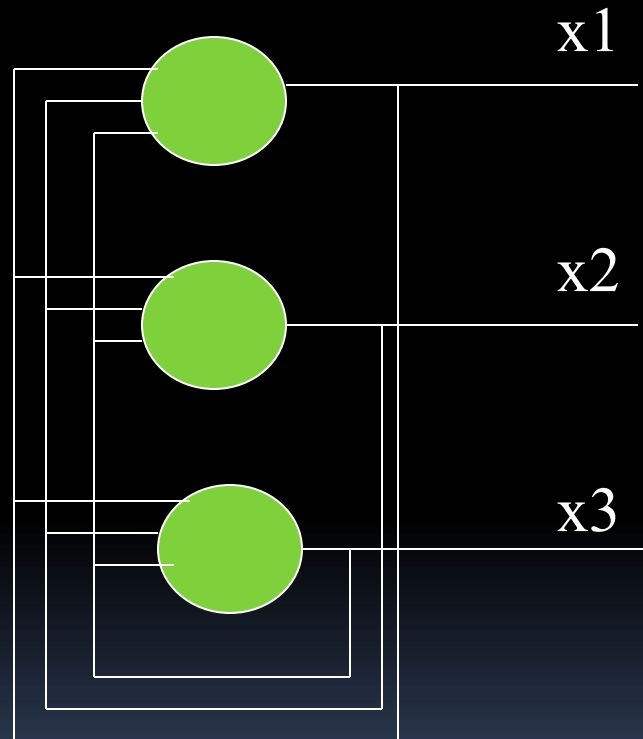




Hopfield network

- Feedback network
 - Easy to train
 - Single layer of neurons
 - Neurons fire in a random sequence
- 

Hopfield network

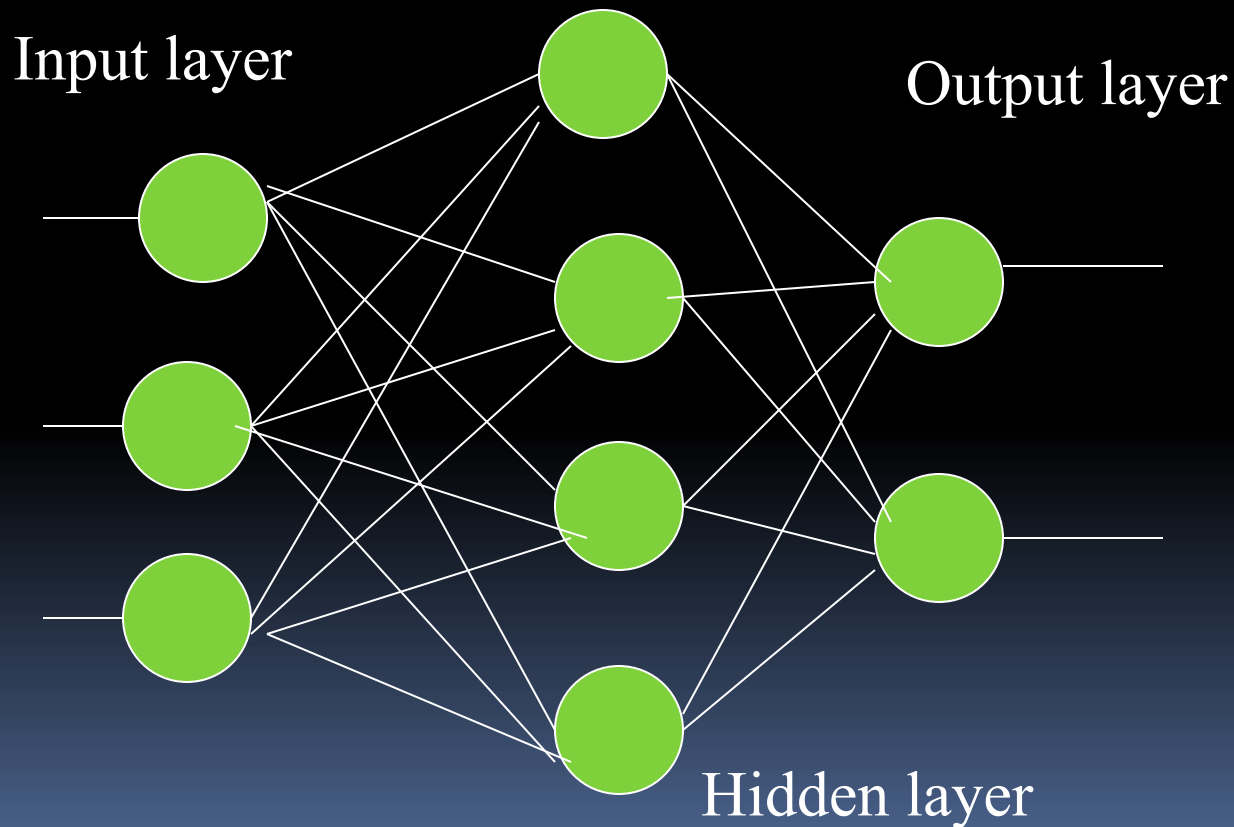




Radial basis function network

- Feedforward network
 - Has 3 layers
 - Hidden layer uses statistical clustering techniques to train
 - Good at pattern recognition
- 

Radial basis function networks



Kohonen network

- All neurons connected to inputs not connected to each other
- Often uses a MLP as an output layer
- Neurons are self-organising
- Trained using “winner-takes all”



What can they do?

- Perform tasks that conventional software cannot do
 - For example, reading text, understanding speech, recognising faces
- 



Neural network approach

- Set up examples of numerals
 - Train a network
 - Done, in a matter of seconds
- 



Learning and generalising

- Neural networks can do this easily because they have the ability to learn and to generalise from examples
- 



Learning and generalising

- Learning is achieved by adjusting the weights
 - Generalisation is achieved because similar patterns will produce an output
- 



Summary

- Neural networks have a long history but are now a major part of computer systems
- 



Summary

- They can perform tasks (not perfectly) that conventional software finds difficult
- 

- 
- Introduced
 - McCulloch-Pitts model and logic
 - Multi-layer perceptrons
 - Hopfield network
 - Kohonen network

- 
- Neural networks can
 - Classify
 - Learn and generalise.
- 